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Who Said It? Speaker-Conditioned Verification and Lexicographic Tie-Breaking for Low-Cost Hallucination Detection in Legislative Hearings

Gabriel Ribeiro^{1,a}, Cauã Sathler^{2,b}, Arthur Gonçalves^{1,b}, Jordan Elias^{3,c}¹Colab Domain-Specific Foundation Models ²Colab Agentic AutoML for Data Science ³Colab SLMs for Process Automation^aDepartamento de Matemática, ^bDepartamento de Ciência da Computação — Universidade Federal de Minas Gerais (UFMG)^cDepartamento de Geografia — Universidade Federal do Ceará (DGEO-UFC)gabriel.ribeiro@dcc.ufmg.br, cauasathler@ufmg.br, afariag72@gmail.com,
jordanelias@alu.ufc.br

Abstract

Abstract. Verifying a summary of a public hearing requires knowing who said what, not merely whether the hearing discussed the topic. We evaluate inexpensive speaker-conditioned signals on PublicHearingBR, using 4,238 generated opinions labelled for support in four retrieved evidence chunks. On the 3,630 opinions with a matched speaker, conditioning cosine similarity on that speaker raises AUROC from 0.568 to 0.746; combining incidence features reaches 0.761 without LLM-judge calls. We then combine these signals with twelve released LLM-judge votes. A logistic model fitted jointly on votes and features falls below the unweighted committee in 19 of 20 hearing-level fold partitions. Lexicographic tie-breaking instead preserves every strict committee ordering, using the secondary score only within equal vote counts. Its exact AUROC gain is the tied-pair fraction times the secondary score’s advantage over chance on those pairs. The full committee improves from 0.926 to 0.930 in all twenty partitions; a direct, untrained misattribution score already reaches 0.9294. Gains are larger with fewer judges: a typical single judge rises from 0.807 to 0.879. In an exploratory review-queue analysis, the best single judge with tie-breaking finds 52.2% of benchmark positives within a 10% review budget, compared with 40.5% without it. These results support inexpensive, traceable triage for human review.

Keywords: Hallucination detection; Large language model judges; Speaker attribution; Lexicographic tie-breaking; Inference cost

1 Introduction

Legislative public hearings are long events with many speakers. When they are summarised, whether by journalists or by language models, each summary sentence usually attributes an opinion to a named participant. An opinion that the participant did not express is a hallucination with public consequences: it attributes words to an identifiable person. Detecting such errors requires comparing each attributed opinion with what that person actually said. PublicHearingBR [4] makes this problem measurable. It contains 206 transcripts of public hearings of the Brazilian Chamber of Deputies, reference summaries, and an NLI evaluation set of 4,238 opinions generated in the dataset authors’ summarisation experiment. A human annotator marked 504 (11.9%) as not inferable from four retrieved chunks, hence *possible* hallucinations rather than verified absence from the entire transcript. We retain these official labels throughout. The dataset also provides the verdicts of twelve LLM judges (three prompts applied to four models); each judge is a language model asked whether the opinion is supported, and it returns a binary vote. We measure detection quality by the Area Under the Receiver Operating Characteristic Curve (AUROC; formal definition in Section 3.1). A single judge reaches an AUROC between 0.734 and 0.847, and the

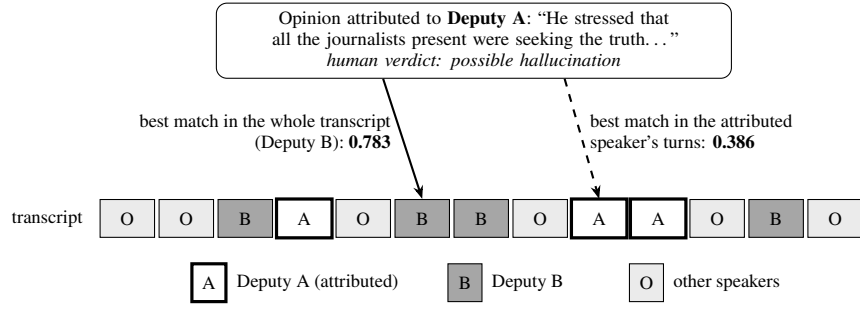


Figure 1. A misattribution from the dataset (participants pseudonymised). The order of turns in the strip is schematic; the two similarity scores are measured. A score computed over the whole transcript finds strong support (0.783), but that support comes from another deputy (Deputy B). Restricted to the attributed speaker’s turns, the best match drops to 0.386.

sum of the twelve votes, which we compute from the released verdicts, reaches 0.926, providing a strong baseline for the comparisons below. Unless stated otherwise, every AUROC in this paper is computed on the 3,630 opinions whose speaker could be identified in the transcript (Section 5). Each judge call has a cost, however, and it is not known how inexpensive signals should be used alongside the judges. Inexpensive detectors compare the opinion with each sentence of the transcript, using embedding similarity or natural language inference (NLI), and keep the best match. On this task they reach an AUROC of only 0.57 (similarity) and 0.58 (entailment) (Section 6), far below the judges, because a plausible hallucination is on topic: the hearing did discuss the subject, so some sentence of the transcript resembles the opinion.

Our starting observation is that the best match may belong to the wrong person. Figure 1 shows a real case, with the participants pseudonymised. In a hearing on press freedom, the summary attributes to Deputy A the opinion “*Destacou que todos os jornalistas presentes estavam em busca da verdade, independentemente de suas preferências políticas*” (“He stressed that all the journalists present were seeking the truth, regardless of their political preferences”), and the human annotator marked it as not supported (a possible hallucination under the annotation protocol). The most similar sentence in the transcript has a cosine similarity of 0.783, but it was spoken by Deputy B. The most similar sentence among Deputy A’s own turns scores 0.386 and concerns a different point. A score computed over the whole transcript sees a well-supported opinion; a score computed over the attributed speaker’s turns sees an unsupported one. This case motivates a structural limitation (Section 6.3): when sentence scores omit speaker identity, any score that combines sentence-level scores through a maximum, a mean or a similar function ignores which speaker produced each sentence, and therefore cannot separate the two situations (Observation 4).

This observation raises two central questions, which organise the paper. The first question is how much conditioning on the speaker helps, and at what cost. We compare the opinion with the attributed speaker’s turns and with the other speakers’ turns, and use these comparisons as features. We call them *speaker-incidence features*, because they use which speaker uttered each sentence. The best match in the attributed speaker’s turns alone raises the AUROC from 0.57 to 0.75, and a logistic regression on all the features reaches 0.76, without LLM-judge calls. The second question is how such an inexpensive detector should be combined with the costly committee of judges. The usual approach fits a model, such as a logistic regression, on the judges’ votes and the new features together. On this dataset it does not improve the committee: its AUROC falls below the committee’s in 19 of 20 random assignments of hearings to folds. The reason is structural: the committee’s vote count takes only thirteen values, so many opinions are tied, and a second detector could usefully order them. A weighted sum, however, does more than order the ties: it can also reverse two opinions that the committee orders correctly, whenever the new detector disagrees strongly enough (Proposition 5). We instead rank opinions by the committee’s vote count and use the new detector only to order opinions that received the same count. This *lexicographic* combination preserves every strict committee ordering, and its gain is given by an exact formula (Proposition 8) that depends on two quantities: how often the committee cannot order a pair of opinions, and how well the new detector orders those pairs. Both can be estimated on a labelled sample before the combination is deployed, and together they indicate where the rule pays off: its gain is largest when judges are few, which is where it saves the most calls.

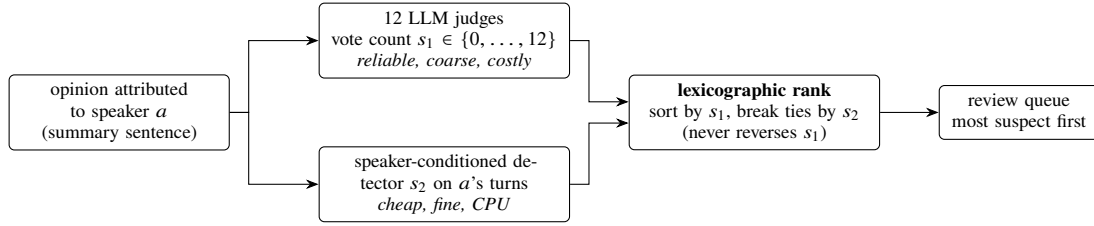


Figure 2. Overview of the method. The detector reads the attributed speaker’s turns and is used only to order opinions the judges cannot order.

Contributions.

- Speaker-incidence features, motivated by the observation that pooled verification scores ignore speaker attribution. They raise AUROC from 0.57 to 0.76 at no LLM cost (measured CPU time in Section 8), and an audit of 50 positives finds that about a quarter were classified as misattributions of this kind (Sections 4.1 and 6).
- A rule for combining a reliable but coarse detector with an inexpensive but finer one. The rule preserves the reliable detector’s strict orderings, and an elementary decomposition of AUROC gives its gain exactly (Section 4.2). Estimated on one half of the hearings, the decomposition forecasts the average gain on the other half using development data alone (Section 7.4).
- A reduction in the number of LLM calls needed for a given accuracy. The rule raises each single judge by 0.046 to 0.101 AUROC, and in an exploratory analysis eight judges with the rule are non-inferior to the full committee of twelve. With the best single judge, a reviewer who checks the 10% most suspect opinions finds 52.2% of the hallucinations instead of 40.5% (Sections 7.4 and 7.6).
- A characterisation of what remains: on the pairs of opinions that the committee orders incorrectly, the inexpensive detector performs below chance, so rules that overrule the committee when the two disagree did not robustly recover the remaining gap in our experiments (Section 7.5).
- A reproducible evaluation including negative results (Section 10) and simple, untrained tie-breaking controls; all code is available.¹

2 Related Work

Hallucination detection and LLM judges. Fernandes et al. [4] introduce the corpus, the human verdicts and the twelve LLM-judge annotations used here. Maynez et al. [7] propose the standard taxonomy of faithfulness errors in abstractive summarisation. The error our features target, content attributed to the wrong speaker, is identified in dialogue summarisation by Ramnath et al. [8], who report that LLM judges detect it only moderately well. Zheng et al. [10] study the reliability of LLM judges, including position, verbosity and self-preference biases. Importantly, the original PublicHearingBR verification pipeline already retrieves four chunks from the attributed person’s speech. Our contribution is to quantify inexpensive incidence signals against pooled baselines and controls, and to use them to refine the released judges’ coarse rankings. The cosine/NLI features and the released judges differ in both evidence selection and scoring; conditioning is isolated by comparing each feature with its own unconditioned counterpart.

Combining detectors. The question of when a classifier should abstain goes back to Chow [2], with the risk–coverage formalism of El-Yaniv and Wiener [3]. A coarse detector abstains implicitly when it assigns the same score to two items, and Lemma 2 quantifies the effect of these ties on AUROC. Ensemble methods usually combine scores by weighted sums; the decomposition of AUROC into tied and separated pairs that we use is elementary, and its value is practical.

3 Background

3.1 AUROC and tied pairs

An *opinion* is a generated summary statement attributed to a named participant. We call the official unsupported labels *positives* or *possible hallucinations*, and the supported labels *negatives*. These names

¹<https://github.com/caua-sathler/IdeiasEmRede-TopoIA>

refer to the four-chunk annotation protocol, not a new judgement of the full transcript. A detector assigns a score s , with larger values indicating greater suspicion.

Definition 1 (AUROC). For independently and uniformly sampled positive and negative opinions $p \in P$, $n \in N$, the Area Under the Receiver Operating Characteristic Curve is the Mann–Whitney statistic [5]:

$$\text{AUROC}(s) = \Pr[s(p) > s(n)] + \frac{1}{2} \Pr[s(p) = s(n)]. \quad (1)$$

AUROC counts correctly ordered positive–negative pairs, giving half credit to ties. Chance is 0.5 and perfect ranking is 1. Unlike accuracy, it cannot reward a detector simply for predicting the majority class, which covers 88.1% of this dataset. It is invariant to strictly increasing transformations of a score. We complement it with recall at fixed review budgets (Section 7.6).

Lemma 2 (Tie accounting). *Let $A(s) = \Pr[s(p) > s(n)]$ and $\tau(s) = \Pr[s(p) = s(n)]$. Then $\text{AUROC}(s) = A(s) + \tau(s)/2 \leq 1 - \tau(s)/2$, with equality exactly when all separated pairs are correctly ordered.*

Proof. Equation (1) gives the identity; $A(s) \leq 1 - \tau(s)$ gives the bound. This standard accounting identity is not a contribution. $\square$

3.2 Pooled scores and rankings with ties

A pooled verifier combines sentence scores $f(o, u_i)$ using a maximum, mean or top- k mean. If those scores omit speaker identity, their aggregate cannot recover it. We write T for the transcript, T_a for the sentences spoken by participant a , and σ for the sentence-to-speaker map.

Definition 3 (Detector preorder). A real-valued score induces a *total preorder* $x \preceq_s y \iff s(x) \leq s(y)$: a reflexive, transitive ranking in which all items are comparable and ties are allowed. Write $x \approx_s y$ for a tie and $x \prec_s y$ for strict order. Its equivalence classes are the groups of equal scores. A refinement preserves all strict orders and may split ties. All preorders below are total and induced by scores.

The primary score s_1 is a judge vote or committee vote count, the secondary score s_2 an inexpensive detector. We use τ for the primary tied-pair fraction and α for the secondary AUROC restricted to those positive–negative pairs. Their product determines the gain of breaking ties, as shown next.

4 Theory

4.1 Pooled scores ignore the speaker

Fix an opinion o attributed to speaker a , a transcript $T = (u_1, \dots, u_m)$ with a speaker map $\sigma : T \rightarrow \mathcal{A}$ that assigns each sentence to the participant who uttered it, and any sentence-level score $f(o, u_i)$.

Observation 4 (Incidence blindness). *Any pooled score $G(\{f(o, u_i)\}_i)$, where G is a maximum, a mean, a top- k mean or any other permutation-invariant function, is invariant under every reassignment of the speaker map σ . Consequently, no such score can distinguish an opinion supported by the turns of its attributed speaker from the same opinion supported only by the turns of other speakers.*

Proof. G does not receive σ as an argument. For a concrete separation, take two sentences with scores 0.9 and 0.1. Under σ_1 the first sentence belongs to a ; under σ_2 it belongs to some $b \neq a$. Every pooled G takes the same value on both configurations, but only the first one supports the opinion as attributed. $\square$

Equivalently, exchanging the speaker labels of all turns in a hearing leaves every pooled score unchanged, although it changes the answer to the question of whether the attributed person said what the opinion claims. No choice of pooling function avoids this. The remedy is to pool over the attributed speaker’s sentences $T_a = \sigma^{-1}(a)$ instead of the whole transcript, and to compare the result with the other speakers’ sentences. Section 6 shows that this restriction raises AUROC by 0.18, and that additional features describing how the support is distributed across speakers add a smaller, significant improvement.

4.2 Combining a coarse and a fine detector

Consider a *reliable but coarse* detector s_1 , such as the committee, which is accurate when it separates two items but assigns the same value to many of them, and an *inexpensive but fine* detector s_2 , which separates almost all items but is less accurate. We call s_1 the **primary** and s_2 the **secondary** detector throughout. The standard way to combine them is a weighted sum, or a logistic regression on both scores, which we call *stacking*.

Proposition 5 (A weighted sum can reverse the primary detector). *Let $w_1, w_2 > 0$ and let $x \prec_1 y$, that is, $s_1(x) < s_1(y)$. The weighted sum reverses this pair; $w_1 s_1(x) + w_2 s_2(x) > w_1 s_1(y) + w_2 s_2(y)$, if and only if $s_2(x) - s_2(y) > (w_1/w_2)(s_1(y) - s_1(x))$. Consequently, a sufficiently large ratio w_2/w_1 reverses some strict relation of $\preceq_1$ if and only if the two detectors disagree on some pair, that is, $s_1(x) < s_1(y)$ and $s_2(x) > s_2(y)$ for some items x, y .*

Proof. The first claim rearranges the inequality. If $s_1(x) < s_1(y)$ and $s_2(x) > s_2(y)$, the condition holds for every ratio $w_2/w_1 > (s_1(y) - s_1(x))/(s_2(x) - s_2(y))$. Conversely, if a pair with $s_1(x) < s_1(y)$ is reversed, the condition gives $s_2(x) > s_2(y)$. $\square$

To correct a pair that the committee orders wrongly, a secondary detector must disagree with the committee on that pair. A weighted sum that gives it enough weight acts on every such disagreement, and therefore also overturns decisions that the committee got right.

Example 6 (A concrete reversal). Consider three opinions. Let s_1 be the committee's vote count divided by twelve and s_2 an inexpensive score in $[0, 1]$, with equal weights.

	x	y	z	
committee s_1	4/12	5/12	5/12	$x \prec_1 y \approx_1 z$
inexpensive score s_2	0.95	0.10	0.90	
sum $s_1 + s_2$	1.28	0.52	1.32	reverses $x \prec_1 y$
lexicographic	(4/12, 0.95)	(5/12, 0.10)	(5/12, 0.90)	keeps $x \prec y \prec z$

The sum ranks x (four votes) above y (five votes) because the inexpensive score disagrees with the committee. The lexicographic order keeps the committee's decision and uses s_2 only to separate y from z , which the committee could not order.

Definition 7 (Lexicographic combination). Given total detector preorders $\preceq_1$ and $\preceq_2$,

$$x \preceq_{\text{lex}} y \iff x \prec_1 y \text{ or } (x \approx_1 y \text{ and } x \preceq_2 y). \quad (2)$$

Operationally: rank items by s_1 , and within each group of items with equal s_1 , rank them by s_2 .

Proposition 8 (Lexicographic combination). $\preceq_{\text{lex}}$ is a total preorder containing $\preceq_1 \cap \preceq_2$, and

- (i) it never reverses a strict relation of $\preceq_1$: if $x \prec_1 y$, then $x \prec_{\text{lex}} y$;
- (ii) it refines $\preceq_1$ only within its equivalence classes, and uses no other information;
- (iii) if α denotes the AUROC of s_2 restricted to the positive–negative pairs that $\preceq_1$ ties (set $\alpha = 1/2$ if $\tau = 0$), then

$$\text{AUROC}(\preceq_{\text{lex}}) = \text{AUROC}(s_1) + \tau(s_1)(\alpha - \frac{1}{2}) \quad \text{exactly.} \quad (3)$$

The combination therefore recovers the fraction $2\alpha - 1$ of the maximum possible gain $\tau/2$. It attains the full $\tau/2$ if and only if $\alpha = 1$, and in all cases $\text{AUROC}(\preceq_{\text{lex}}) \in [\text{AUROC}(s_1) - \tau/2, \text{AUROC}(s_1) + \tau/2]$.

In words, the gain equals the share of tied pairs times how much better than a coin flip the secondary detector orders them.

Proof. Totality and transitivity of (2) follow directly from the definition. For the inclusion, let $x \preceq_1 y$ and $x \preceq_2 y$: either $x \prec_1 y$ and the first clause applies, or $x \approx_1 y$ and the second applies. Claim (i) is the first clause, and claim (ii) the second. For (iii), let $A = A(s_1)$, so that $\text{AUROC}(s_1) = A + \tau/2$ by Lemma 2. By (i), the pairs that s_1 separates are ordered in the same way by $\preceq_{\text{lex}}$ and contribute A . By (ii), the tied pairs, of total probability τ , are ordered by s_2 alone; applying Definition 1 to this sub-population, they contribute $\tau\alpha$ instead of $\tau/2$. Hence $\text{AUROC}(\preceq_{\text{lex}}) = A + \tau\alpha$, and subtracting $\text{AUROC}(s_1)$ gives (3). The bound follows from $\alpha \in [0, 1]$. $\square$

Remark 9 (The gain depends on α , not on the global AUROC of s_2). By Proposition 8(i), the lexicographic combination keeps every pair that s_1 separates in the order s_1 gives it. The pairs that s_1 orders wrongly, which account for the gap $1 - \tau/2 - \text{AUROC}(s_1)$, are therefore out of its reach, and for any secondary detector the change in AUROC lies within $\pm\tau/2$. Within this range, Equation (3) involves only two quantities: τ , which depends on the primary detector and the labels but not on s_2 , and α , the AUROC of

s_2 on the tied pairs. The AUROC of s_2 on the whole task does not appear. Two practical consequences follow. First, a candidate secondary detector should be evaluated on the primary detector’s ties rather than on the whole task: our features reach 0.78 overall but only $\alpha = 0.59$ on the committee’s ties. Second, the maximum gain $\tau/2$ can be computed before any secondary detector is built, and once α is estimated on a development set the gain follows without running the combination. Section 7.4 tests this forecast on unseen hearings.

Remark 10 (From the lexicographic combination to the weighted sum). Let $r_1(x) \in \{1, \dots, K\}$ index the primary score’s classes, and $r_2(x) \in [0, 1]$ be the secondary score’s normalised rank. The score $f_\gamma = r_1 + \gamma r_2$ induces the lexicographic order for every $0 < \gamma < 1$: the secondary term cannot bridge the unit gap between primary classes. At $\gamma = 1$, extreme adjacent items may tie; for $\gamma > 1$, some primary orders may reverse. Thus lexicographic tie-breaking can also be implemented as a bounded additive perturbation, without selecting a weight on the evaluation labels. Section 7.5 examines whether allowing reversals helps empirically.

5 Data and Methods

This section describes the data, the statistical protocol shared by all experiments, the speaker-incidence features, and the two detectors combined in Section 7.

Data. We use all 206 hearings of PublicHearingBR: 4,238 generated opinions with human NLI annotations (504 positive, 11.9%) and stored votes from four models with three prompts each. Both human and LLM verdicts concern the four chunks retrieved by the original speaker-conditioned pipeline. Our features instead access the transcript sentences and recovered speaker turns. Sentences and opinions are embedded with `paraphrase-multilingual-mpnet-base-v2` [9]. A *turn* is a block of consecutive sentences spoken by one participant. Turns are recovered from the transcript’s speaker markers (0 SR. NAME -): each sentence is assigned to the most recent marker. The participant named in the summary is matched to a turn speaker by overlap of name tokens. The match succeeds for 85.7% of the opinions ($n = 3,630, 408$ positive), which form the main evaluation set, since both the speaker-incidence features and the combination experiments require the attributed speaker’s turns. A failed match is itself informative and is used as a feature in Section 6. Entailment probabilities come from `mDeBERTa-v3-base-xnli` [6], a natural language inference (NLI) model that scores whether a premise entails a hypothesis; each sentence pair evaluated is one *NLI call*.

Statistical protocol. All learned models are evaluated *out of fold*: we use five-fold cross-validation in which all opinions of a hearing fall in the same fold (`GroupKFold`), so each opinion is scored by a model fitted on other hearings and no model is evaluated on data it was fitted on. Models are logistic regressions with balanced class weights and $C = 1$; features are standardised using only each training fold. Confidence intervals resample whole hearings rather than opinions, since opinions from the same transcript are not independent (3,000 to 5,000 bootstrap resamples, depending on the analysis; all intervals are 95% and, when two detectors are compared, paired on the same resamples). Two additional sources of variance are controlled in the combination experiments. A random tie-break is itself a random variable, so we compare against 200 random draws. The assignment of hearings to folds is not captured by the bootstrap, so the main combination comparison is repeated over 20 random assignments of hearings to folds (*fold partitions*) and reported as median and range. Where a single value is reported, it comes from the deterministic five-fold reference partition. Feature-only models and combination models use the explicitly specified feature sets below. We report negative findings alongside positive ones; research hypotheses and their provenance are archived in `code/predictions/`.

Speaker-incidence features. For an opinion o attributed to the matched speaker a , let $S(o, \cdot)$ be the cosine similarity between o and each transcript sentence, T the set of all sentences, and T_a the set of sentences spoken by a .

$\cos_g$	$\max_{s \in T} S(o, s)$: best match in the whole transcript (baseline)
$\cos_s$	$\max_{s \in T_a} S(o, s)$: best match in the attributed speaker's turns
Δ	$\max_{b \neq a} \max_{s \in T_b} S(o, s) - \cos_s$: how much better another speaker supports the opinion
share_a	sum of the similarities of the ten best global matches that lie in T_a , divided by their total
rank_a	rank of a among all speakers ordered by their best match, scaled to $[0, 1]$
cover_a	1 if a is among the strong supporters, i.e. the speakers with at least one sentence of similarity ≥ 0.65
$n_{\text{sup}}, n_{\text{op}}$	number of speakers with similarity ≥ 0.65 ; number of opinions attributed to the matched speaker in the hearing
$\text{nli}_s, \text{nli}_g$	maximum entailment probability over the five best-matching sentences of a / of the whole transcript

The feature $\cos_s$ asks whether the attributed speaker's own turns support the opinion, and Δ asks whether somebody else supports it better; we call Δ the *misattribution score*. When a feature is used alone as a detector, it is oriented so that larger values indicate a more suspect opinion (for example, $-\cos_s$). The remaining features describe how much of the support belongs to the attributed speaker, where that speaker ranks, and whether the speaker is among the supporters at all. None of these quantities can be computed from a pooled score (Observation 4). The cosine features require no model call beyond the precomputed embeddings; the two entailment features require ten NLI calls per opinion. NLI uses the top five cosine matches in each pool, with premise and hypothesis truncated to 280 characters and paired inputs to 192 tokens. The cosine regression uses $(\cos_s, \text{best}_{\text{other}}, \text{share}_a, n_{\text{sup}}, \text{cover}_a, \text{rank}_a)$; the NLI variant adds $(\text{nli}_s, \text{nli}_g, \text{nli}_g - \text{nli}_s)$. Combination experiments additionally include n_{op} , explaining the small difference from feature-only AUROCs.

The two detectors combined in Section 7. The primary detector $s_1 \in \{0, \dots, 12\}$ is the number of judges that flag an opinion. This unweighted vote count outperforms logistic weights fitted out of fold on the same votes (0.9260 vs. 0.9245 on the representative partition, and a median of 0.9226 over twenty partitions, Table 3). The secondary detector is a logistic regression on all the speaker-incidence features, including NLI and n_{op} (0.7787, up to ten NLI calls per opinion); unless stated otherwise, Section 7 uses this detector. *Stacking* denotes a logistic regression fitted out of fold on the votes and the same features together. As a control, Section 7.7 also uses a deliberately weak inexpensive detector, $\cos_g$.

6 Results I: Speaker-Incidence Features

6.1 Conditioning on the attributed speaker

Table 1 reports the main result of this section. On the 3,630 matched opinions, the best match in the whole transcript reaches an AUROC of 0.5676 (0.5628 over all 4,238 opinions). Computing the same similarity over the attributed speaker's turns raises it to 0.7455, a difference of +0.178 (clustered CI [+0.128, +0.237]).

Table 1. AUROC of each signal against the human verdict on the matched-speaker subset ($n = 3,630$, 408 positives). “0 calls” means the signal is computed from precomputed embeddings and speaker markers alone.

Signal	Cost / opinion	AUROC
$\cos_g$: best match in the whole transcript (baseline)	0 calls	0.5676
nli_g : best entailment in the whole transcript (baseline)	up to 5 NLI	0.5828
$\cos_s$: best match in the attributed speaker's turns	0 calls	0.7455
Δ : misattribution score	0 calls	0.7595
share_a : share of the support	0 calls	0.7135
rank_a : rank of the speaker	0 calls	0.7258
cover_a : speaker among strong supporters	0 calls	0.6706
all speaker-incidence features, out of fold	0 calls	0.7605
all features + $\text{nli}_s, \text{nli}_g$	~10 NLI	0.7792

Three controls address alternative explanations. **Speaker identity.** Computing the same feature over a *randomly chosen other* speaker of the same hearing, matched for number of turns, gives 0.5209,

which is at chance and below the global baseline. The paired difference between the attributed and the random speaker is $+0.2246$ [$+0.1903$, $+0.2592$], so the signal comes from *who* spoke, not from how much. **Number of sentences.** The number of sentences of the attributed speaker alone scores 0.4953 (correlation $+0.26$ with $\cos_s$), so the effect is not an artefact of taking a maximum over fewer sentences. **Features beyond the single score.** Out of fold, the full set of features improves on the best single conditioned score, from 0.7447 to 0.7605, a paired difference of $+0.0159$ [$+0.0028$, $+0.0298$]. Additionally, two complementary findings emerge from the data: first, a failed speaker match is informative (hallucination rate 15.8% vs. 11.2%), and using it as an indicator feature extends the detector to all 4,238 opinions (0.7435 out of fold); second, conditioning on the speaker also benefits entailment, where nli_s improves on nli_g by $+0.1264$ [$+0.0928$, $+0.1614$]. Contrary to our hypothesis, however, nli_s alone (0.7095) does not outperform $\cos_s$ (0.7455); entailment adds value only in combination with the cosine features ($+0.0186$ [$+0.0060$, $+0.0309$]).

6.2 Comparison and combination with individual judges

The twelve judges each emit a binary vote and reach an AUROC between 0.7340 and 0.8468. Without LLM-judge calls, the speaker-incidence features (0.7605) outperform the three weakest judges (0.7340, 0.7401, 0.7522). Combined with any single judge by *rank sum*, which adds the two ranks and requires no fitting, they improve that judge in twelve cases out of twelve, with every clustered CI excluding zero. The gains range from $+0.027$ to $+0.094$, are largest for the weakest judges (per-judge results are in the repository), and the best such rank-sum combination reaches 0.8823. Section 7.4 reports the lexicographic combination, which performs better. Section 7 examines the full twelve-judge committee and how to combine the features with it without degrading it.

Interpretation. One type of error these features detect has a simple description: a plausible hallucination that is on topic for the hearing but not for the turns of the person to whom it is attributed. A global similarity score cannot see this difference (Observation 4), whereas the speaker-conditioned features are built to capture it. That the strongest single feature is the misattribution score Δ (0.7595) is consistent with this interpretation. The example of Figure 1 (0.783 in the whole transcript, 0.386 in the attributed speaker’s turns, $\Delta = +0.40$) is one such case: its misattribution score exceeds that of 99.3% of the grounded opinions, in agreement with the human verdict. Section 6.3 measures how common this type of error is among the positives.

6.3 What kind of error are the positives?

We examined a random sample of 50 of the 408 benchmark positives (seed 20260923) with two Claude annotation passes, using retrieved speaker-specific and global evidence with neighbouring sentences. The model annotations agree on 45 items (Cohen’s $\kappa = 0.81$). Both assign 12 items (24%) to misattribution, 30 to apparent support by the attributed speaker and 3 to distortion; five disagreements are left unresolved. All 12 consensus misattributions have positive Δ (mean 0.21). Apparent support outside the official retrieval context is consistent with the benchmark’s four-chunk scope. This exploratory audit characterises the error mechanisms without replacing the official labels. Sampling, category assignments and the protocol are provided in `code/AUDIT_PROTOCOL.md`.

7 Results II: Combining the Features with the Committee

7.1 How the committee’s ties are distributed

The sum of the twelve votes takes thirteen values and reaches an AUROC of 0.9260. The ranking it induces is nevertheless coarse: $\tau = 4.65\%$ of positive–negative pairs are tied, so by Lemma 2 the largest possible gain from ordering them is $\tau/2 = 0.0232$, and a perfect secondary detector would reach 0.9492 (not the bound $1 - \tau/2 = 0.9768$ of Lemma 2, which would also require the committee to order all separated pairs correctly). This does not mean that the judges are close to a performance ceiling: their sensitivity ranges from 0.48 to 0.92 and their specificity from 0.64 to 0.99, and two of them miss more than half of the hallucinations while lying within 0.009 of their own bound $1 - \tau/2$. All gains reported below are bounded by $\tau/2$, not by that larger gap.

Figure 3 explains why the tie fraction is small although half of the opinions receive zero votes. The tie fraction is a sum of *products* of class shares. The zero-vote class holds 58% of the grounded opinions but

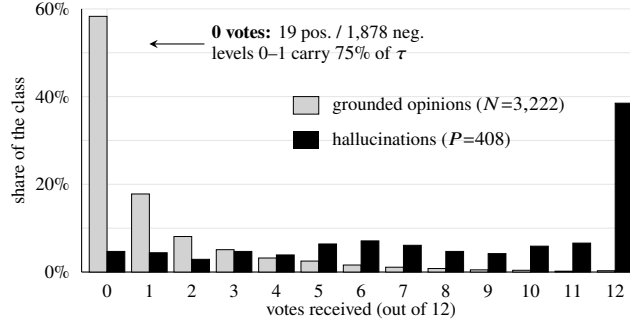


Figure 3. The committee’s tie structure. For each of the thirteen vote levels, the share of grounded opinions (grey) and of hallucinations (black) receiving that many votes. The tie fraction is $\tau = \sum_k \pi_k^+ \pi_k^-$, the sum of products of the two bars at each level: level 0 alone contributes 58% of it ($\pi_0^- = 0.583$, $\pi_0^+ = 0.047$), levels 0–1 contribute 75%.

only 4.7% of the hallucinations (19 against 1,878 opinions). Hallucinations rarely receive zero votes; they concentrate at twelve votes (38.5%), where almost no grounded opinion is found.

7.2 Decomposing the effect of each combination

Because AUROC is an average over pairs (1), it can be decomposed exactly over any partition of the pairs. We divide the 1,314,576 positive–negative pairs into those the committee *separates* (95.4%) and those it *ties* (4.6%), and measure the effect of each combination on each group (Table 2).

Table 2. Exact decomposition of AUROC over pair types. “Reversals” counts separated pairs whose order the fused score flips: *harmful* if the committee had them right, *benign* if wrong. One representative fold partition; stability is Table 3.

Method	AUROC	contribution from		reversals among separated pairs		
		separated	tied	total	harmful	benign
committee (vote count)	0.9260	0.9027	0.0232	—	—	—
stacking (logistic)	0.9262	0.9013	0.0249	1.33%	0.74%	0.59%
lexicographic	0.9300	0.9027	0.0272	0.00%	0.00%	0.00%

Tied pairs contribute exactly $\tau/2 = 0.0232$ to the committee’s AUROC, as Lemma 2 requires. The weighted combination (“stacking”) extracts information from them, raising their contribution to 0.0249 (+0.0017), but it also reverses 1.33% of the separated pairs, which costs 0.0014, so the net gain is 0.0002. The lexicographic combination reverses no pair and raises the contribution of the tied pairs to 0.0272 (+0.0040).

7.3 Main comparison

Table 3 evaluates the combinations across twenty independent GroupKFold partitions, supporting two main conclusions. First, stacking is worse than the committee in median, and it is unstable: its result varies by 0.0087 depending only on the fold assignment, a range that includes the committee’s value, so a single run may report either a gain or a loss. Second, the lexicographic combination varies by 0.0012, seven times less, and always exceeds the committee in all partitions. Table 4 gives the paired comparisons on the representative partition.

Table 3. AUROC over twenty independent GroupKFold partitions. The reference, the committee’s vote count (0.9260), does not depend on the folds. “Features” denotes the speaker-incidence features, including NLI. The last column gives the share of partitions in which the combination exceeds the committee.

Combination	median	min	max	> committee
stacking (committee + features)	0.9247	0.9189	0.9276	5%
committee, out-of-fold weights	0.9226	0.9175	0.9245	0%
lexicographic (committee, features)	0.9301	0.9295	0.9307	100%

Table 4. Paired AUROC differences on the representative fold partition (95% hearing-clustered bootstrap intervals, paired resamples).

Comparison	Δ AUROC	95% CI
lexicographic vs. committee (vote count)	+0.0040	[+0.0005, +0.0078]
lexicographic vs. committee (OOF weights)	+0.0054	[+0.0005, +0.0105]
lexicographic vs. stacking	+0.0038	[+0.0001, +0.0078]

How much does an untrained score recover? As an exploratory simplicity control, we also break committee ties directly by $-\cos_s$ or Δ , without regression or NLI (Table 5). Direct Δ reaches 0.9294 and improves on the committee by 0.0034 (paired hearing-bootstrap CI [+0.0004, +0.0068]). The full detector adds 0.0006 over this control (CI [-0.0019, +0.0031]); its additional benefit is not established here. Thus an interpretable score without training already captures much of the committee-level gain. The full detector remains the common secondary score in the budget and review-queue experiments below.

Table 5. Exploratory controls for the twelve-judge committee. All rows preserve its strict ordering. Paired intervals use 3,000 hearing resamples; script `simple_tiebreak_baselines.py`.

Secondary score	Trained	NLI pairs/opinion	AUROC
none (committee alone)	no	0	0.9260
$-\cos_s$	no	0	0.9291
Δ	no	0	0.9294
full feature regression	out of fold	up to 10	0.9300

7.4 The gain at each level of cost

Table 6 applies the lexicographic combination to committees of 1, 3, 5, 8 and 12 judges. The rule improves every committee size, and its gain decreases as the committee grows, as Equation (3) implies: larger committees tie fewer pairs. The gain is largest at a single judge call. Every one of the twelve judges improves, by +0.046 to +0.101, with all twelve hearing-clustered intervals excluding zero; a typical judge rises from 0.807 to 0.879, and recovers about 60% of the difference between one judge and the full committee. The additional cost is ten calls to a small NLI model; without them the typical gain is still +0.068. With more judges the medians in Table 6 show the same pattern: five judges with the rule (0.923) are above eight without it (0.921).

Table 6. AUROC by number of LLM judge calls per opinion. For k judges, values are medians over 60 random subsets of k judges (the single full committee for $k = 12$) on one fold partition. “Alone” is the vote count of the subset; the next two columns add lexicographic tie-breaking with the speaker-incidence features, without and with the ten NLI calls. In the first row, “alone” is the global-cosine baseline and the features are used as the detector. The gain is that of the last column over the first. The regression additionally includes n_{op} , unlike Table 1. Script `judge_budget.py`.

Judge calls	with tie-breaking			gain
	alone	0 NLI calls	10 NLI calls	
none	0.5676	0.7611	0.7787	+0.2111
1	0.8067	0.8750	0.8791	+0.0724
3	0.8917	0.9096	0.9114	+0.0197
5	0.9128	0.9209	0.9226	+0.0098
8	0.9205	0.9253	0.9271	+0.0066
12	0.9260	0.9289	0.9300	+0.0040

Forecasting the gain on unseen hearings. We test a fully prospective use of Equation (3): split the hearings in half, fit the secondary detector only on development hearings, and estimate both τ and α there (the latter using out-of-fold predictions). The forecast $\tau_{dev}(\alpha_{dev} - 1/2)$ then uses neither test labels nor test scores. We compare it with the realised gain on the other half over 50 random splits and five committee sizes (Table 7). Mean forecasts are 92%–98% of mean observed gains; their signs agree in 249 of 250 cases. Individual forecasts are less precise: at twelve judges the MAE is 0.0033 for a mean observed gain of 0.0042. Replacing conditional α by the development-set global AUROC systematically overestimates

the mean gain. The equation is therefore useful for estimating average benefit, rather than guaranteeing the gain of a particular deployment.

Table 7. Fully prospective gain forecasts: both τ and α (or global AUROC) are estimated on development hearings only. Means and mean absolute errors (MAE) over 50 splits per committee size. Script `gain_forecast.py`, column `pred_dev`.

Judges	observed gain	forecast with α		with global AUROC	
		mean	MAE	mean	MAE
1	+0.0721	+0.0705	0.0123	+0.0997	0.0283
3	+0.0284	+0.0263	0.0091	+0.0478	0.0193
5	+0.0152	+0.0147	0.0067	+0.0309	0.0158
8	+0.0075	+0.0073	0.0047	+0.0185	0.0111
12	+0.0042	+0.0040	0.0033	+0.0128	0.0086

The gain expressed in judge calls. Across twenty fold partitions and sixty random subsets per size, eight judges with tie-breaking reach a median AUROC of 0.9272, versus 0.9260 for all twelve. They match or exceed the full committee in 73% of the evaluated configurations; for nine judges, the proportion is 97%. For an eight-judge subset selected to have median mean performance across partitions, the paired bootstrap comparison on the first partition gives +0.0008 (CI $[-0.0031, +0.0048]$). This is a different statistic from the across-configuration median. Its lower bound exceeds the exploratory non-inferiority margin -0.0040 , chosen to match the full-committee gain. With seven judges the corresponding lower bound is -0.0046 . These retrospective comparisons suggest potential savings of three to four calls; they do not establish a prospective judge-selection policy. Choice of judges matters more than fold assignment (standard deviations 0.0025 and 0.0004, respectively, at seven judges).

Individual judges. Applied to each judge separately, the lexicographic combination outperforms stacking for all twelve judges (nine with clustered CIs excluding zero), and the largest gains go to the judges with the most ties. The headline single-call figures are those of a typical judge (Table 6). For the best judge (`p2_gpt4omini`, 0.8468), a variant selected after the fact, the combination reaches 0.8928, against 0.8877 with stacking.

Where the combination is useful. Both τ and α depend on how coarse the primary detector is. The best single judge (AUROC 0.8468) ties $\tau = 26.1\%$ of the pairs, a maximum gain of 0.1307, which is 5.6 times that of the twelve-judge committee. On these ties the speaker-incidence features reach $\alpha = 0.676$, against 0.5856 on the committee’s ties. Equation (3) accounts exactly for the observed gains, $+0.0460$ ($0.8468 \rightarrow 0.8928$) and $+0.0040$ ($0.9260 \rightarrow 0.9300$).

7.5 How much of the possible gain is obtained, and why not more

Equation (3) expresses the observed gain as a property of the secondary detector. On the representative partition, the speaker-incidence features reach $\alpha = 0.5856$ on the committee’s ties (median 0.589 over twenty partitions), so the combination recovers $2\alpha - 1 \approx 17\%$ of the maximum gain $\tau/2$. The value $\tau(\alpha - \frac{1}{2})$ equals the observed $+0.0040$; we verified the identity numerically to six decimal places. Table 8 shows how this value arises.

Table 8. AUROC of the speaker-incidence features on each group of pairs, by what the committee does with the pair; medians over twenty fold partitions. The middle row is the maximum gain of Proposition 8(iii). The last row limits everything else: the features have little aggregate corrective signal where the committee is wrong.

the committee...	% of pairs	contributes	features’ AUROC
separates, correctly	90.3%	0.9027	0.809
ties	4.6%	0.0232	0.589
separates, wrongly	5.1%	0.0000	0.473

The maximum gain is highly concentrated (Figure 3). Three quarters of it depends on the 37 hallucinations (19 + 18) that received zero or one vote. These are, by construction, the hardest cases in the

dataset, since eleven or twelve judges did not flag them, and only 19 of them lie in the zero-vote class, against 1,878 grounded opinions. On these cases the features reach $\alpha = 0.56$; where they perform well (0.87 at twelve votes), only 2.3% of the maximum gain is available.

Limits of the remaining gap. As Table 8 shows, the features perform below chance (0.473) on the pairs that the committee orders incorrectly, and this value never exceeds 0.482 across twenty partitions. This shows little aggregate corrective signal on those pairs, but does not exclude useful subgroups or other models. The tested combinations did not robustly close this gap: only 44% of the pairs reversed by stacking were initially wrong (Table 2).

We evaluated six further attempts to raise α (fitting the secondary detector within each committee class, adding the vote count as a feature, interactions with the vote count, interactions with four coarse strata, weighting by the mass of tied pairs, and a pairwise ranking loss on the tied pairs only) all *lower* it, from a median of 0.589 to between 0.489 and 0.520, in all twenty fold partitions. A gradient-boosted-tree configuration on votes and features also reaches only 0.9001. Within these experiments, keeping the secondary detector separate from the votes proved preferable. With the vote count as a feature, the secondary detector’s global AUROC rises to 0.9260, because it relearns the committee, while its α falls to 0.514. The norm of its speaker-incidence coefficients halves ($0.921 \rightarrow 0.470$) and their direction rotates (cosine 0.889). Since the vote count is constant within a tied class, only this rotated and weaker direction remains to order the ties.

Finally, the combinations f_γ of Remark 10, which allow the secondary detector to move an item across up to γ classes of the committee, do not help with the actual features: $\gamma = 2$ exceeds the lexicographic combination by only +0.0005 (CI $[-0.0004, +0.0013]$) and exceeds its median in 85% of the twenty partitions, while giving up the guarantee of Proposition 8(i). The mechanism is not the limitation: with a perfect secondary detector, f_γ ranges from 0.9492 to 0.9975 as γ grows. To estimate how good a new feature would need to be, we replaced the secondary detector by the true label, flipped at random for a fraction ρ of the opinions, and for each ρ between 0 and 0.5 searched for a value of γ at which f_γ outperforms the lexicographic combination. Overruling the committee helps only while the detector’s AUROC on the committee’s incorrectly ordered pairs exceeds approximately 0.58.² The speaker-incidence features reach 0.473 on those pairs.

7.6 Effect on a human review queue

AUROC evaluates how a detector orders pairs of opinions. An auditor uses the order differently: opinions are reviewed from the most to the least suspect until a fixed budget is spent, and what matters is the share of hallucinations found within that budget. A gain in AUROC changes this share only if it reorders opinions near the budget boundary. Figure 4 reports, for each detector, the share of the 408 benchmark positives found when the top $k\%$ of opinions are sent to a human reviewer.

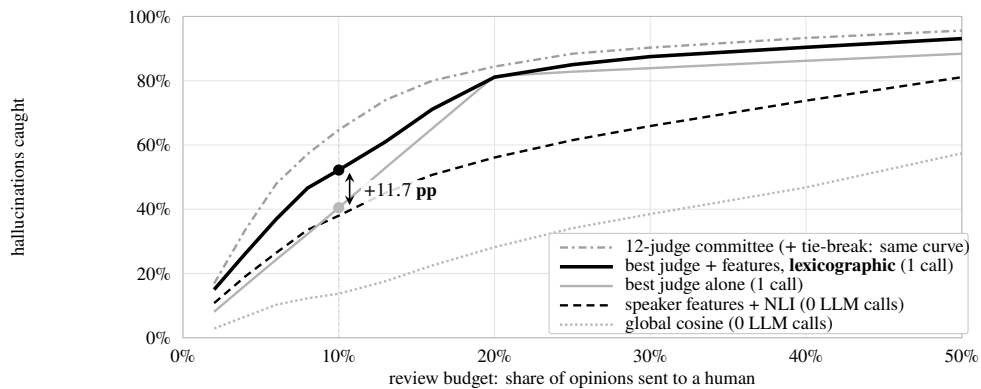


Figure 4. Share of hallucinations found as a function of the review budget. The curve of the best judge alone is a straight line up to 20%: a binary verdict does not order the opinions within each of its two classes, so the queue is in random order there. Ordering these ties with the speaker-incidence features adds 11.7 percentage points at a 10% budget with one judge call; with twelve judges the difference is not measurable.

²Script `gamma_sweep.py`; the value is interpolated between the tested levels of ρ .

At the inexpensive levels, the operational gain is large and agrees with the theory. When one opinion in ten is reviewed, the global-cosine baseline finds 13.7% of the hallucinations, the speaker-incidence features 38.0% without LLM-judge calls (with local NLI), and a single judge combined with them 52.2%, against 40.5% for the same judge alone. This single-judge gap follows from Equation (3) with $\tau = 26.1\%$: a binary judge ties most pairs, so most of its queue is unordered.

With twelve judges, however, the gain in the queue is not measurable. We hypothesised that the lexicographic queue would outperform the committee’s at every review budget. It does not: the differences are -0.2 , $+0.3$ and $+0.6$ percentage points at budgets of 5%, 10% and 20%, and every hearing-clustered interval includes zero. The location of the committee’s ties explains why. Three quarters of the tied pairs involve opinions with zero or one vote (Figure 3), and these opinions occupy the last 69% of the committee’s queue: ordering them raises the AUROC but does not change which opinions are reviewed under any budget below 30%. Within such a budget only the vote level at the boundary is tied, and even a perfect ordering of that level could change the recall by at most 1.7 to 4.7 points. At twelve judges the result is therefore a ranking improvement, not a workload improvement. For the selected single judge it improves both metrics, because the best judge’s single tied class of flagged opinions spans the first 20% of its queue.

7.7 Controls

Ordering the ties at random has the same expected AUROC as not ordering them, but a non-negligible variance (over 200 draws: mean 0.9259, standard deviation 0.0020, maximum 0.9308); against this distribution the lexicographic result of 0.9300 has $p = 0.015$. A deliberately weak secondary detector, the global cosine ($\alpha = 0.5409$), gives a non-significant $+0.0021$ $[-0.0017, +0.0057]$, as its low α predicts. The committee must come first: reversing the order would lose the guarantee of Proposition 8(i).

8 Measured Cost

“Low cost” needs evidence. We measured CPU time on a machine with 4 vCPUs and no GPU, and the tokens a judge reads (XLM-R tokenizer as a proxy; Table 9). We did not measure monetary cost, since provider prices change.

Table 9. Measured cost. The last two rows are per judge call; the judges’ fixed prompt text is not included.

Component	Cost
speaker parsing, name matching and incidence features	0.17 ms / opinion
embedding the transcript (once per hearing; 982 sentences on average)	38.9 s / hearing
amortised over the 20.6 opinions of a hearing	≈ 1.9 s / opinion
NLI, ten sentence pairs	≈ 3.2 s / opinion
tokens read by one judge (4 chunks + opinion)	600 (median), 875 (p95)
tokens of a whole transcript, for contrast (mean, 40 hearings)	27 283

The cosine features cost about 1.9 s of CPU per opinion, dominated by embedding the transcript (paid once per hearing); the speaker-incidence computation itself takes under a millisecond. The measured components of the full secondary detector, including ten NLI pairs, total about 5.1 s per opinion. These component benchmarks exclude opinion embedding and retrieval/index overhead; they are not end-to-end latency. Neither makes an LLM call, whereas each judge call reads about 600 tokens: twelve judges read about 7,200 tokens per opinion and the eight judges of our non-inferiority result about 4,800.

9 Ethics, Data Use and Limits of Use

Data. We use only PublicHearingBR, built from official public transcripts of the Brazilian Chamber of Deputies [1], under the terms of the challenge, of the Câmara dos Deputados and of Hugging Face. No other personal data is collected; names are pseudonymised in the figures and text, and no per-person score is reported.

Limits of use. The detector flags *opinions* for human review; it does not rank or profile individuals, and a flag is not evidence that a person did not express a view, since the labels are an upper bound on hallucination. An opinion should not be attributed to, or withdrawn from, a named person on the detector’s output alone.

10 Threats to Validity

Negative results. We report findings that contradicted our hypotheses. Conditioned entailment does not outperform conditioned cosine similarity (Section 6); at twelve judges the combination does not improve the review queue (Section 7.6); opinions whose supporting sentences contradict one another are not more often unsupported (AUROC 0.43 to 0.46); and fitting the secondary detector within each committee class lowers α , as do the five other attempts of Section 7.5.

Labels. We retain the released four-chunk NLI labels [4]; they may overestimate full-transcript hallucination. The 206 hearings form one corpus, so transfer to other domains or summarisation systems remains to be evaluated. The audit of Section 6.3 characterises the positives but does not replace the labels.

Binary judges. The maximum gain $\tau/2$ would shrink if the judges produced calibrated probabilities; our combination result is conditional on binary votes.

Speaker recovery. Speakers are matched by name for 85.7% of the opinions, and matching errors are not necessarily conservative: excluding hearings with sign-language interpretation (1.1% of the opinions) lowers the main signal from 0.7455 to 0.7314. All entailment probabilities come from a single multilingual NLI model.

Scope of the claims. The gain over the full committee is small (+0.0040, clustered CI lower bound +0.0005) and rests on its consistency: it holds in all twenty fold partitions and for all twelve individual judges (the full-committee comparison has $p = 0.015$ against random tie-breaking); the advantage over stacking (+0.0038, lower bound +0.0001) is of the same size, and stacking is itself unstable across fold partitions. The primary comparisons are the three of Section 7.3; the per-judge results, the decomposition of Section 7.5 and the audit are exploratory.

11 Conclusion

Pooled verification scores cannot tell whether an opinion was expressed by the person to whom it is attributed. On PublicHearingBR, conditioning on the attributed speaker’s turns raises the AUROC of an inexpensive detector from 0.57 to 0.76 without LLM-judge calls. Adding such a detector to a committee of LLM judges, however, requires care: fitting a single model on both can overturn the committee’s strict ordering, and on this dataset it falls below the committee in 19 of 20 fold partitions. In contrast, ranking by the committee and using the inexpensive detector only to order its ties never reverses a strict committee ordering, and its gain equals the tie fraction times the detector’s advantage over chance on the tied pairs. Both quantities can be estimated on a small labelled sample, allowing average benefit to be estimated before deployment, as we verified on held-out hearings. The tested alternatives did not robustly close the remaining gap, and the features provide little aggregate signal on pairs the committee orders incorrectly.

Practical implications. The main consequence is a reduction in the number of LLM calls needed for a given accuracy. One judge call combined with the rule raises a typical judge from 0.81 to 0.88, recovering about 60% of the difference to the full committee with one twelfth of its LLM calls; with the best judge, a reviewer who checks the 10% most suspect opinions finds 52.2% of benchmark positives instead of 40.5%. Exploratory subset comparisons suggest that eight judges with the rule can approach twelve-judge performance, potentially saving a third of calls. For a new corpus, one can measure the tie fraction of the available judges on a labelled sample, estimate the detector’s AUROC on the tied pairs, and adopt the rule where the forecast gain $\tau(\alpha - \frac{1}{2})$ justifies it. Negative findings delimit when these gains translate into a better review queue.

Reproducibility. All numbers in this paper are produced by the scripts in the code/ directory of the public repository (<https://github.com/caua-sathler/IdeiasEmRede-TopoIA>). Analyses use fixed random seeds and precomputed tables; feature extraction and CPU remeasurement have separate commands. A bilingual interactive dashboard and a review-queue demonstration accompany the code. We thank Instituto Kunumi for the *Ideias em Rede* challenge and the authors of PublicHearingBR.

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